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Living Causal KAN

A Kolmogorov-Arnold causal graph that updates online

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Design proposal · November 2026 · not yet implemented or tested

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Outline

1. **The causal graph**: a KAN whose nodes are real variables;
2. **Splines**: degree, knots, B-spline basis, coefficients;
3. **Updates**: evidence, score, decision, commit;
4. **Reading an update**;
5. **The overall architecture**, in progress.

Notation

The graph and its curves

x_j	Value of variable j for one person
i, j	A parent and its child
d	Number of variables
e	One edge
ϕ	A learned curve on one edge (small phi)
u_j	Noise: what the parents do not explain
σ	Sigmoid: turns any number into a chance
pa_Y	The parents of Y
k, G	Spline degree, and number of intervals
ξ_m	Knot number m
B_m	Bump number m (one B-spline)
θ_m	Coefficient: the weight of bump m

The updates

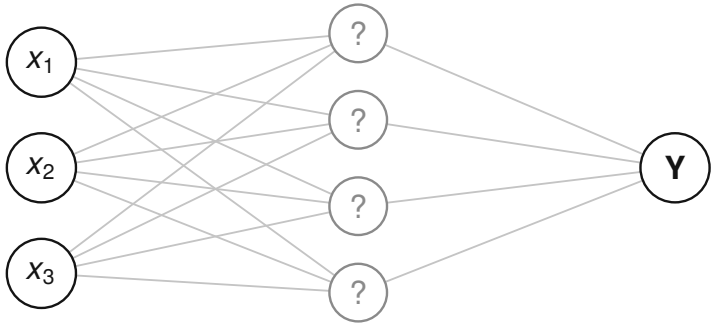
Φ_j	List of bump values for one sample (capital phi)
$\Phi_j^\top \theta_j$	Prediction: multiply pairs, then add
A_j, b_j	Running statistics of node j
γ	Share of old evidence kept at each sample
t, r	The current sample, and a past sample
s_j, s_e	Score of node j, and its part on edge e
$\|\|s_e\|\|$	Size of a list as one number: $\|\|[3,4]\|\| = 5$
δ	An edit to the coefficients
λ, λ_2	Price of an edit: per edge, per coefficient
S	The coefficients chosen for an edit
y, p	Observed outcome, and predicted chance of $Y=1$
$\leftarrow$	Replace the old value with this one

In ϕ^t, the t is a version label, not a power. In γ^{t-r} it is a power: older samples count less.

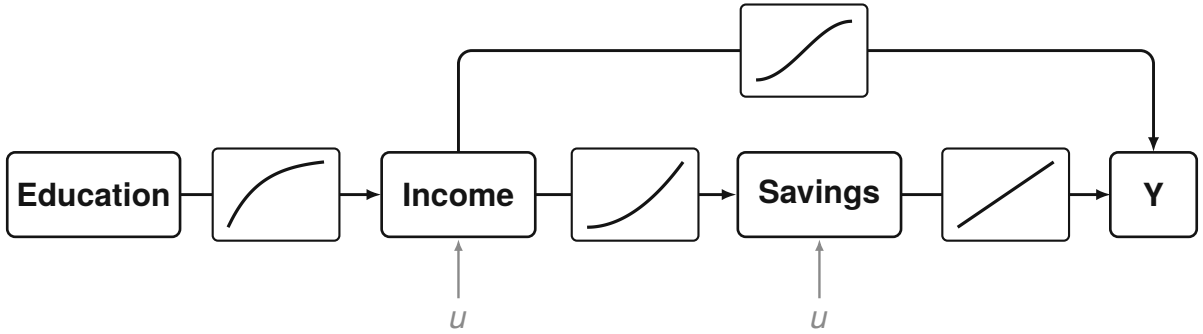
Design question · causal graph

Can the causal graph be a KAN?

A standard KAN



The causal graph as a KAN



	Standard KAN	Causal graph as a KAN
Hidden nodes	Unnamed and unmeasured	Real variables you measure
Wiring	Every node to the next layer	Causal edges only, and they can skip a layer
Training	All at once, to predict Y	Each node on its own measured column
Action do(a)	Not defined	Set a node, then recompute what follows it

Choice: Yes. Make the graph the KAN, with measured variables as its nodes, so an action has a meaning.

A KAN has learned curves on edges and sums at nodes (Liu et al. 2024), which is also the form of an additive causal model.

Design question · causal graph

Why one curve per parent, and no hidden layers?

One curve per edge: a node adds the curves of its parents, so Y, with two parents, has two curves.

Inside a node	Interactions	Drift on one edge	Online update
One curve per parent (this design)	None	Yes	Exact, from running statistics
Hidden layers, as in a standard KAN	All	No	Needs gradient training

Readable

Each causal edge has its own curve, so a drift lands on one edge.

Online

Each feature equation stays linear in its coefficients, so fixed-size statistics are enough.

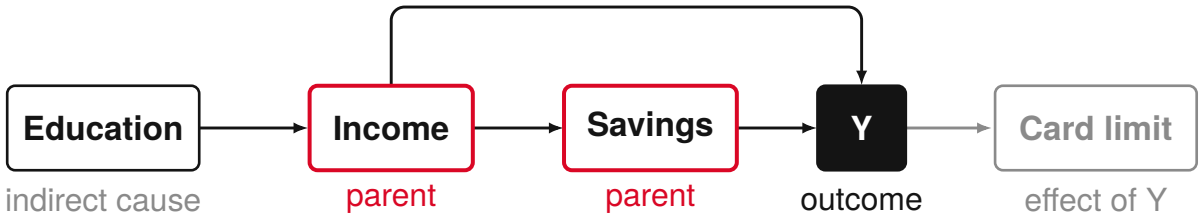
Still deep

Curves compose along causal paths: education reaches Y through income.

Choice: One curve per parent, added at the node.

Design question · causal graph

What is Y, and what makes a variable its parent?



$$P(Y{=}1 \mid \mathrm{pa}_Y) = \sigma\big(\phi_{\text{income}}(\text{income}) + \phi_{\text{savings}}(\text{savings})\big)$$

The Y equation: the chance of the favourable value, given Y's parents.

Y is what really happens: the person repays ($Y{=}1$, the favourable value) or does not ($Y{=}0$). The bank's decision only predicts it.

A parent is a direct cause: change it with everything else held fixed, and Y's chance still moves.

In the model, a parent is an edge whose curve is not zero.

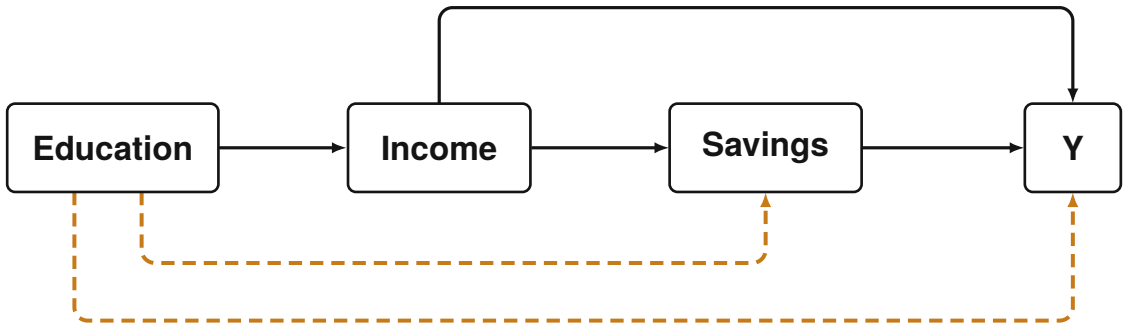
	Example	Changing it changes Y?	Its curve into Y
Parent	Income, savings	Yes, directly	Not zero
Indirect cause	Education	Yes, through income	Zero
Effect of Y	Card limit at other banks	No	None: it comes after Y in the order

Choice: Y is one more node. Its parents are the edges with a non-zero curve, and the order keeps effects out.

Illustrative example.

The causal graph: one equation per variable

Feature equations	$x_j = \sum_{i<j} \phi_{ij}^t(x_i) + u_j$	How variables move each other
Y equation	$P(Y{=}1 \mid \mathrm{pa}_Y) = \sigma\big(\sum_{i\in \mathrm{pa}_Y} \phi_{iY}^t(x_i)\big)$	How Y's parents drive the outcome
Versioned curves	$\phi_e^t = \phi_e^{t-1} + \delta_e^t$	Any past version can be rebuilt



Kept: curve not zero (4) Candidate: curve at zero (2)

Setup, once before the stream

Order: set from domain knowledge. A variable can only receive edges from earlier ones, so d variables give $d(d-1)/2$ candidate edges.

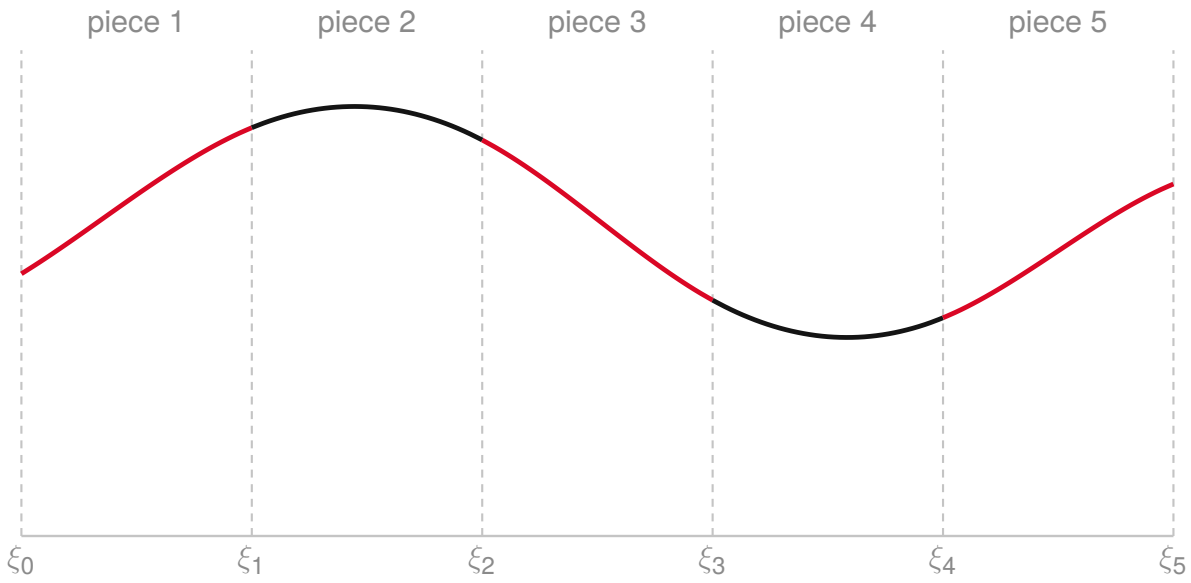
Fit: a sparse fit on warm-up data sends curves with no evidence to zero.

Afterwards: zero curves stay as candidates, so an edge can appear during the stream.

Built on CAM (Bühlmann et al. 2014) and KaCGM (Almodóvar et al. 2026). Not novel, except sparse versions of spline coefficients (adapted).

What a spline is

A spline of degree k is a curve of polynomial pieces of degree k, joined smoothly at knots.



Interval

The stretch between two neighbouring knots. The curve has G intervals.

Piece

One polynomial per interval: $c_0 + c_1 x + c_2 x^2 + c_3 x^3$ for $k = 3$.

Join

At a knot, neighbouring pieces match in value and in derivatives up to order $k - 1$.

Definition: de Boor 1978; Hastie et al. 2009, ch. 5. Example with $k = 3$ and $G = 5$.

Polynomial degree: how far each piece can bend

One basic bump of each degree, between knots

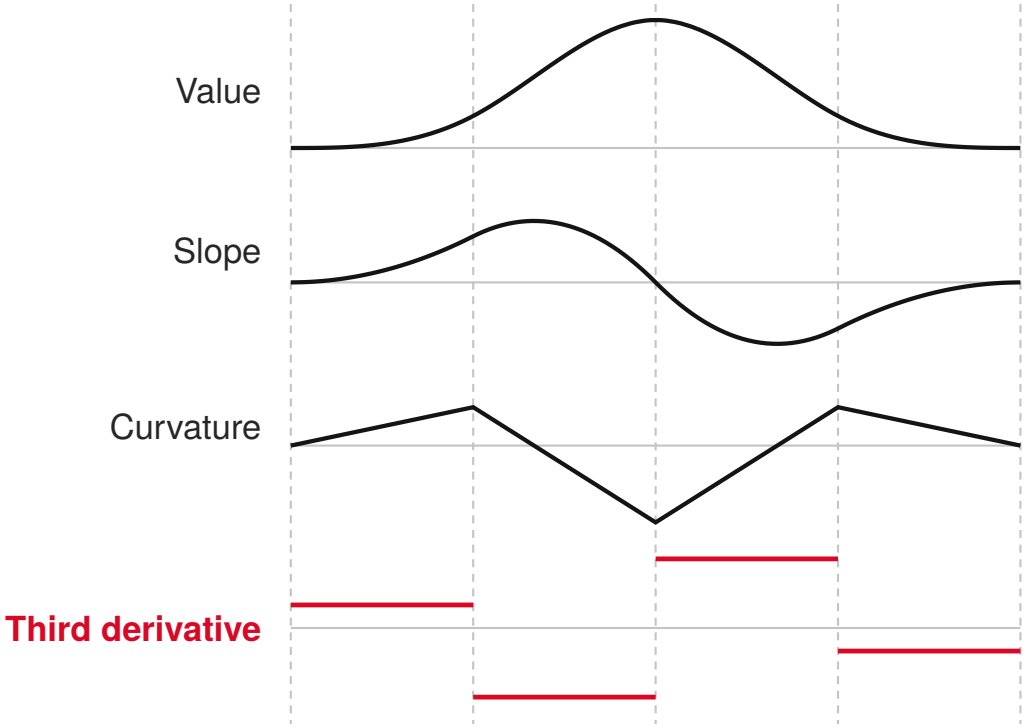


Degree k	0	1	2	**3 (cubic)**
Order, $k+1$	1	2	3	4
Each piece	Constant	Line	Parabola	Cubic
Continuous at a knot	Nothing	Value	Value, slope	Value, slope, curvature
Intervals per bump	1	2	3	4
Active bumps per input	1	2	3	4

Choice: Degree 3. Slope and curvature are continuous at every knot, and each input activates 4 bumps at most.

Terms differ: "order" is $k+1$ in de Boor 1978 and Hastie et al. 2009, but Liu et al. 2024 write "order k" for degree k. Both use cubic splines.

A knot is where the polynomial may change



What it is

A fixed point on the input axis, such as income = 1,500.

What it does

The pieces on each side may differ. For $k = 3$, only the third derivative jumps there.

What it is not

Not a data point, and not learned here. A coefficient is not the curve's height at a knot.

Types

2 boundary knots, $G - 1$ interior knots, and k extra knots past each end to complete the basis.

More knots in a region give the curve more freedom there. Plot: one cubic B-spline and its derivatives.

How many knots are needed?

Counting

G intervals have $G - 1$ interior knots. Each piece has $k + 1$ coefficients, and each interior knot removes k:

$$G(k + 1) - (G - 1)k = G + k \text{ coefficients}$$

Intervals G	Interior knots	Coefficients, $k = 3$
5	4	8
8	7	11
10	9	13

No fixed number: a trade-off

Too few: the curve misses the shape. For smooth functions the error falls like $G^{-(k+1)}$. Too many: fewer samples per coefficient, and a larger A.

Reference points

Restricted cubic splines: 3 to 7 knots, usually 5
CAM: 10 basis functions per variable
KaCGM: grid sizes 1, 5 and 10
P-splines: many knots, plus a smoothness penalty

Choice: G between 5 and 10, a setting to tune. With 500 samples of memory and $G = 8$, each interval keeps about 60 samples.

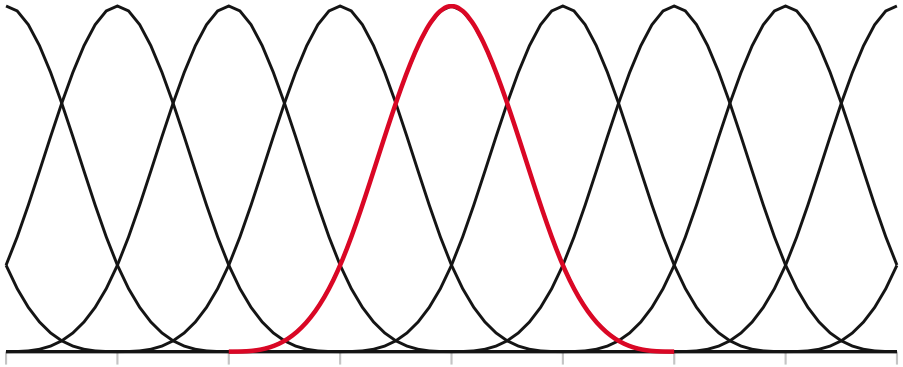
Error rate: Liu et al. 2024, Theorem 2.1. Harrell 2015; Bühlmann et al. 2014; Almodóvar et al. 2026; Eilers & Marx 1996.

B-splines: a basis that is smooth by construction

$B_{m,0}(x) = 1$ if $\xi_m \leq x < \xi_{m+1}$, otherwise 0

$$B_{m,k}(x) = \frac{x - \xi_m}{\xi_{m+k} - \xi_m} B_{m,k-1}(x) + \frac{\xi_{m+k+1} - x}{\xi_{m+k+1} - \xi_{m+1}} B_{m+1,k-1}(x)$$

Each degree is built from two bumps of the degree below. ξ_m are the knots.



No join conditions

Any coefficients give a valid spline.

Local

Each bump is zero outside $k + 1$ intervals.

Sparse

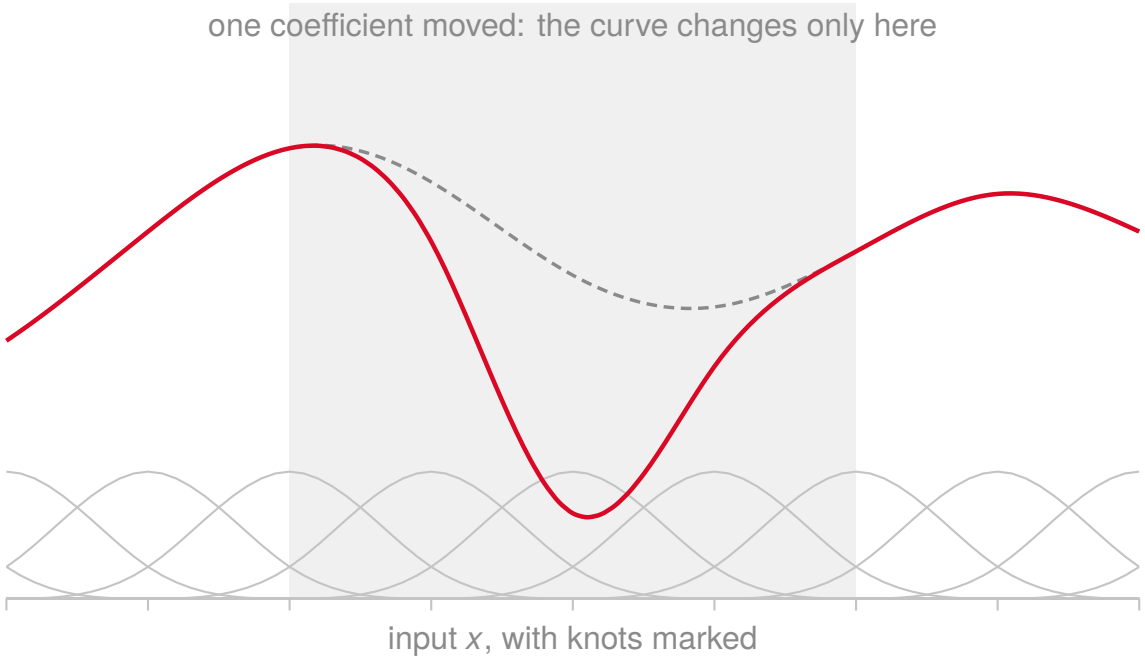
At most $k + 1$ bumps are active at any input.

Sum to one

Bumps are non-negative and add up to 1.

Cox-de Boor recursion: de Boor 1978. Plot: the 11 cubic B-splines for $G = 8$ with uniform knots, with one in red.

A curve is a list of coefficients



1 A weighted sum of bumps

$\phi(x) = \sum_m \theta_m B_m(x)$. The coefficients θ_m are what is learned.

2 One coefficient, one region

Changing θ_m moves the curve only over $k + 1$ intervals.

3 A weight, not a height

The curve follows the θ_m but does not pass through them.

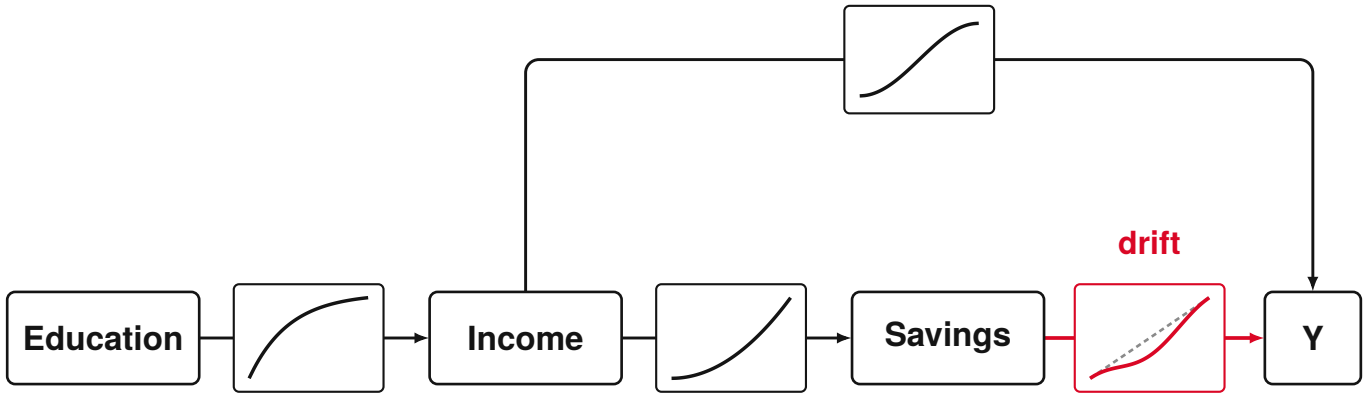
So each node is linear in its coefficients: $x_j = \Phi_j^\top \theta_j + u_j$, where Φ_j holds the bump values of all candidate parents.

Spline settings in this design

Setting	Value	Reason
Degree	$k = 3$	Continuous curvature, 4 active bumps per input
Intervals	$G = 5$ to 10	Enough shape with a small, fixed A
Interior knots	Warm-up quantiles	Equal data per interval at setup
After setup	Knots stay fixed	Old and new coefficients stay comparable
Outside the range	Straight line with the end slope	Polynomials diverge there, and a line stays linear in θ
Constant	One per node, one bump dropped per edge	A constant cannot move between edges
Coefficients	$G + 2$ per edge, 1 per node	$G = 8$ with 2 parents: 21 coefficients

Quantile knots: Harrell 2015. Straight tails, as in natural splines: Hastie et al. 2009, ch. 5.

An update is a small edit to one curve



Each small box is one learned curve on one causal edge.
Red: the only edge that changed. Dashed: its old curve.

1 One curve per causal edge

Each edge is a spline, so its effect can be plotted.

2 Small, sparse edits

$\phi_e^t = \phi_e^{t-1} + \delta_e^t$, and δ_e stays exactly zero without strong evidence.

3 Recorded as a Change

Each edit is saved with its edge, knot range and version.

Small edits assume that few mechanisms change at once. This is the sparse mechanism shift hypothesis, not a fact (Perry et al. 2022).

Update step 1 of 4 · every sample

Collect the evidence

$$A_j \leftarrow \gamma A_j + \Phi_j \Phi_j^\top \qquad b_j \leftarrow \gamma b_j + \Phi_j x_j$$

A: what was seen

One entry per pair of coefficients: how often recent samples activated both.

b: what was observed

Each sample's bump values, weighted by the value observed.

γ: how long it lasts

Memory is about $1/(1-\gamma)$ samples: $\gamma = 0.998$ keeps about 500.

Why this is enough

The least-squares fit is $\hat{\theta}_j = A_j^{-1} b_j$, so raw data is not stored.

Fixed cost

Size depends on the number of coefficients, not on the length of the stream.

The curves do not move in this step.

Exact for feature nodes, which are linear in their coefficients.

Update step 2 of 4 · every sample

Read the score

$s_j = b_j - A_j\theta_j = \sum_r \gamma^{t-r}\,\Phi_r\,(x_{j,r} - \Phi_r^\top \theta_j)$ r is a past sample. One block s_e per edge.

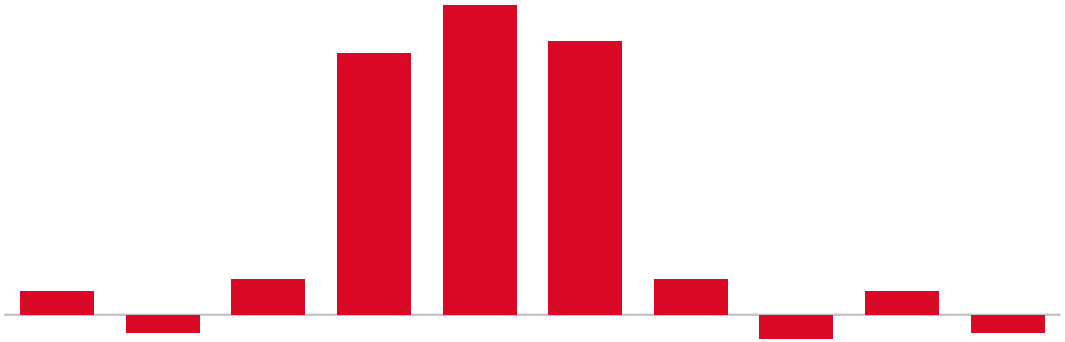
Curve still right

Errors have no pattern and cancel.



Curve wrong in one range

Errors share a sign there and add up.



Small $\|s_e\|$: the sample ends here. Most samples do.

Schematic: the score of each coefficient on one edge.

Update step 3 of 4 · when a score passes the threshold

Decide: one rule detects and updates

$$\min_\delta \ \tfrac{1}{2}\delta^\top A\delta - \delta^\top s + \lambda \sum_e \|\delta_e\| + \lambda_2 \|\delta\|_1$$

Fit: $\frac{1}{2}\delta^\top A\delta - \delta^\top s$

Fall in squared error, from A and s alone.

Edge penalty: $\lambda \sum_e \|\delta_e\|$

Paid per edge, so most edges stay at zero.

Local penalty: $\lambda_2 \|\delta\|_1$

Paid per coefficient, so the edit stays local.

$\delta = 0$ is optimal exactly while every edge has $\|s_e\| \le \lambda$.

The change test is the fit's optimality condition, so no separate drift detector is needed. Below λ, neither noise nor a small real change moves a curve.

Strictly, $\|s_e\|$ with the mean error removed and each coefficient score shrunk by λ_2.
Group lasso (Yuan & Lin 2006), sparse-group lasso (Simon et al. 2013), online sparse estimation (Angelosante et al. 2010).
Per-edge use in an online causal model: possibly novel.

Update step 4 of 4 · only after a decision

Commit

1

Keep the chosen

The set S of coefficients the fit left non-zero.

2

Refit them

$\delta_S = A_{SS}^{-1} s_S$, with no penalty, because penalties shrink the edit.

3

Save a version

$\phi_e^t = \phi_e^{t-1} + \delta_e^t$, so any past curve can be rebuilt.

4

Record a Change

Edge, knot range, δ, version and time.

On S the score falls to zero, because the new curve fits the statistics. A and b are unchanged.

At the first commit the statistics still hold old data, so one real change can take several commits on the same edge.

Refitting after selection: Belloni & Chernozhukov 2013. Sparse versions of spline coefficients: adapted.

Forgetting fades the evidence, not the curves

	Curves	Statistics A, b
Hold	Everything committed so far	Recent evidence
Fade?	Never	By γ on every sample
Change	Only at a commit	On every sample

One factor

The same γ scales A and b, so forgetting alone leaves $A^{-1}b$ unchanged.

A region with no new data

Its evidence fades and its score stays near zero, so δ stays zero and the curve stays.

The Y node: same steps, different statistics

	Feature node	Y node
Equation	$x_j = \Phi^\top \theta + u_j$	$p = P(Y{=}1) = \sigma(\Phi^\top \theta)$
Error on one sample	$x_j - \Phi^\top \theta$	$y - p$
Score	$s \leftarrow \gamma s + \Phi\,(x_j - \Phi^\top \theta)$	$s \leftarrow \gamma s + \Phi\,(y - p)$
A	$A \leftarrow \gamma A + \Phi\Phi^\top$	$A \leftarrow \gamma A + p(1-p)\,\Phi\Phi^\top$
Exact?	Yes	No: p comes from the curves at that moment
When	As the sample arrives	When the outcome arrives, often much later

Open: outcomes are seen only for accepted people, and the approximation has no proof yet.

Design question · updates

Can the graph tell which kind of drift happened?

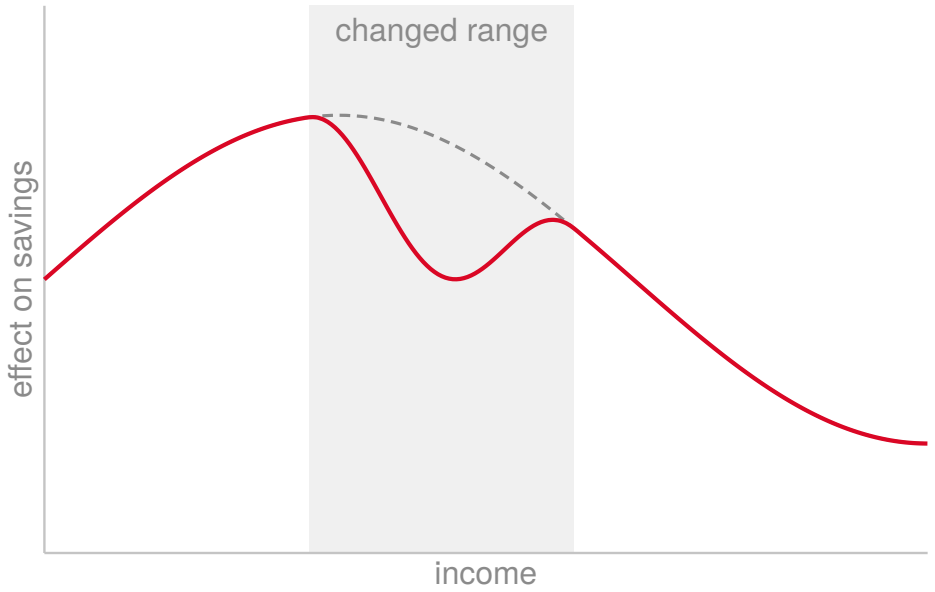
Drift	Example	What the graph shows
Exogenous	Incomes rise for everyone	No curve moves, so no Change
Endogenous	Income adds less to savings	A Change on income → savings
Target	Savings matter less for Y	A Change on savings → Y
Structural	A new direct cause appears	A zero curve leaves zero
Confounder	A hidden cause shifts	Can look like a change on an edge

Choice: **Read it from where the curve moved, on every edge.**

Drift types: Barboza et al. 2026. The mapping onto this design is new and untested.
A reversed edge cannot be shown, because the variable order is fixed.

Design question · updates

Does explaining drift need a second model?



Which edge and where

Each Change names them: only that edge got a δ.

How much

δ is the gap between the old and new curve.

One episode

Commits in a row on one edge are read as one change.

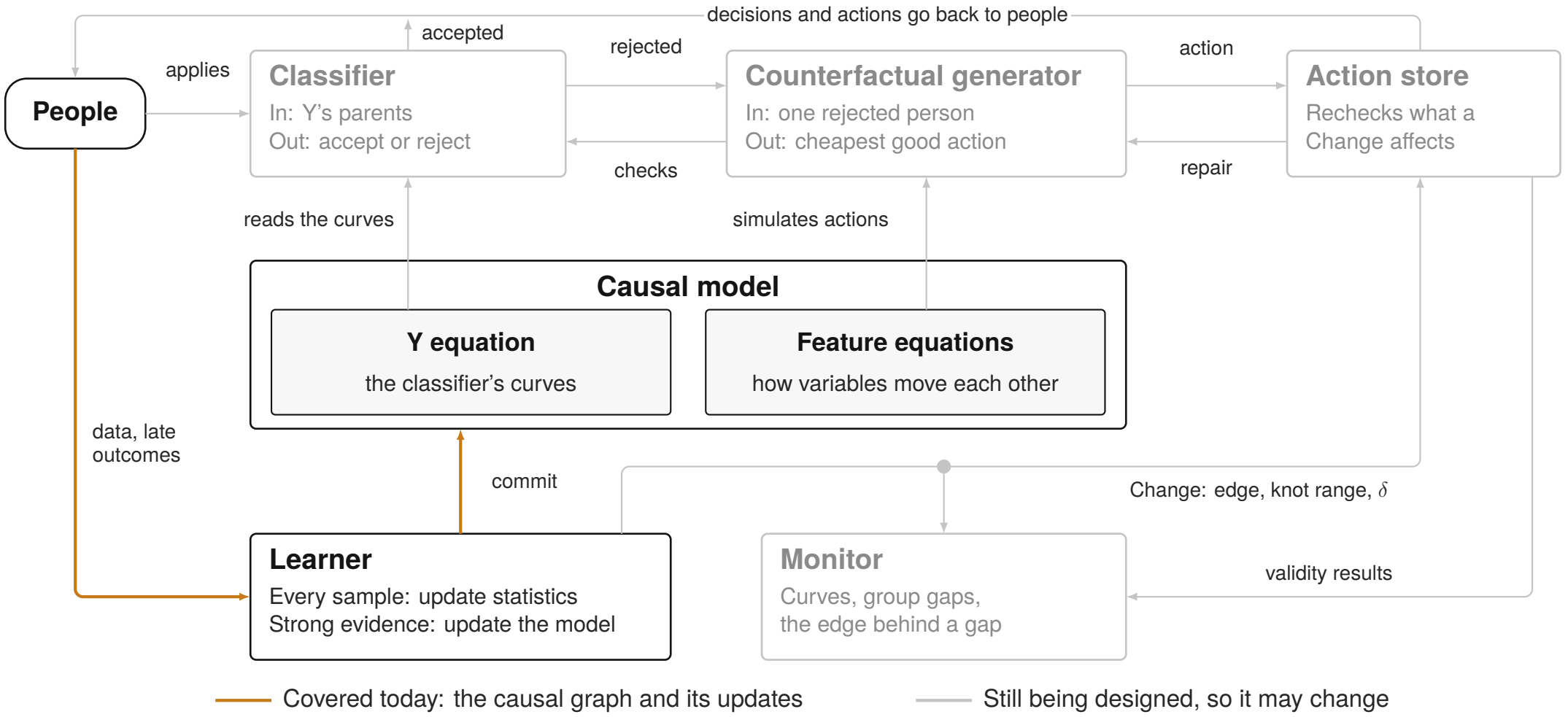
“Between 0.3 and 0.6 of income, the effect on savings is now lower.”

Choice: No. The update is the explanation: you read the curves the model uses.

Schematic. This is the model’s view, so an interaction it cannot represent can give a clean but wrong sentence.

Work in progress

The overall architecture



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† Not read directly: seen as an abstract, a record or through secondary sources. Check before citing in the thesis.